# Brendan Clark

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## Education

**Ph.D.**, Atmospheric Science 2025
Rutgers University, New Brunswick, NJ
"Stratospheric sulfate aerosol climate intervention implications for global agriculture"

**M.S.**, Atmospheric Science, GPA: 3.90/4.00 2022
Rutgers University, New Brunswick, NJ

**B.S.**, Environmental Science 2020
University of Massachusetts, Amherst, MA

## Professional Appointments

Postdoctoral Research Associate, Supervisor: Prof. Daniele Visioni 2025-present
Cornell University, Department of Earth and Atmospheric Sciences

Research Assistant, Supervisors: Professors Alan Robock and Lili Xia 2020-2025
Rutgers University, Department of Environmental Science

## Publications

N. Grant, A. Robock, L. Xia, J. Kiniry, **B. Clark**. Can solar climate intervention save coffee and chocolate from climate change? *In review*

**B. Clark**, D. Visioni., S. Kothari, M. Lerdau. Land models likely underestimate the impact of atmospheric dryness on European tree growth, *In review*

**B. Clark**, A. Robock, L. Xia, S. S. Rabin, J. R. Guarin, J. Jaegermeyr. Stratospheric aerosol climate intervention could reduce crop nutritional value. *Environmental Research Letters*, 20, 114083 (2025). 10.1088/1748-9326/ae1151

**B. Clark**, A. Robock, L. Xia, S. S. Rabin, J. R. Guarin, G. Hoogenboom, J. Jaegermeyr. Maize yield changes under sulfate aerosol climate intervention using three global gridded crop models. *Earth's Future* 13, e2024EF005269 (2025). 10.1029/2024EF005269

N. Grant, A. Robock, L. Xia, J. Singh, **B. Clark**. Impacts on Indian agriculture due to stratospheric aerosol intervention using agroclimatic indices. *Earth's Future* 13, e2024EF005262 (2025). 10.1029/2024EF005262

**B. Clark**, L. Xia, A. Robock, S. Tilmes, J. H. Richter, D. Visioni, S. S. Rabin. Optimal climate intervention scenarios for crop production vary by nation. *Nature Food* 4, 902–911 (2023). 10.1038/s43016-023-00853-3

F. Bowlick, **B. Clark**, et al. Understanding geocomputation education: A survey and syllabi informed review. *Research in Geographic Education* 23, 20-51 (2022).

## Funding and Awards

- Climate Intervention Network Travel Award ($1,000) 2025
- NSF NCAR Exploratory Allocation Award (500,000 CPU hours) 2025
- GeoMIP Annual Meeting Travel Funding Award ($3,500) 2025

- Rutgers Atmospheric Science Student Travel Support Award ($1,500) 2024
- Rutgers Atmospheric Science Student Travel Support Award ($1,200) 2023
- Rutgers Climate Institute Student Travel Support Award ($1,000) 2022
- NSF NCAR University Small Allocation Award (200,000 CPU hours) 2022
- Governor's Award in Recognition of Environmental Stewardship 2019
- John and Abigail Adams Scholarship Award ($7,000) 2016
- Stanley Z. Koplik Certificate of Mastery with Distinction Award 2016

## Teaching & Mentorship Experience

- Guest Lecturer, Rutgers Honors Program Seminar Spring 2026
- Graduate Student Mentor, Cornell University 2025-present
- Guest Lecturer, Climate Modeling, Rutgers University Fall 2023
- Teaching Assistant, Geographic Information Systems, UMass Amherst Fall 2019
- Substitute Teacher, Algonquin Regional High School, Northborough, MA 2018-2019

## First Author Presentations

**Talks**

CMIP Community Workshop (Kyoto, 2026), American Geophysical Union (New Orleans, 2025), Gordon Research Seminar on Climate Engineering, *invited* (Italy, 2024), American Geohysical Union (San Francisco, 2023), Agricultural Modeling Intercomparison Project Meeting (Columbia University, 2023), Solar Climate Intervention Virtual Symposium, *invited* (University of Exeter, 2023), Climate Engineering in Context (University of Potsdam, 2022), NCAR Land Modeling Working Group Meeting (Boulder, 2022), Inter-sectoral Impact Model Intercomparison Project Meeting (University of Potsdam, 2022)

**Posters**

Geoengineering Modeling Intercomparison Project Meeting (Tokyo, 2026), CMIP Community Workshop (Kyoto, 2026), American Geophysical Union (New Orleans, 2025), DEGREES Global Forum (South Africa, 2025), American Geophysical Union (D.C., 2024), Geoengineering Modeling Intercomparison Project Meeting (Cornell, 2024), Gordon Research Conference on Climate Engineering (Italy, 2024), GeoMIP Meeting (University of Exeter, 2023), Gordon Research Conference on Climate Engineering (Maine, 2022), American Geophysical Union (Chicago, 2022), American Geophysical Union (New Orleans, 2021)

## Service

- Refereeing: Atmospheric Chemistry and Physics, JGR - Atmospheres, Environmental Research: Climate
- Physicists Coalition for Nuclear Threat Reduction 2020-present
- Science Policy and Advocacy at Rutgers 2020-2024

## Participation in International Experiments

- Agricultural Modeling Intercomparison Project 2020-present
- Global Gridded Crop Modeling Intercomparison Project 2020-present
- Geoengineering Modeling Intercomparison Project 2020-present
- Climate Intervention Biology Working Group 2020-present
- Inter-Sectoral Impact Model Intercomparison Project 2020-present